

Replication Report

Learning Sequential Distribution System Restoration
via Graph-Reinforcement Learning

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1 Overview

This report documents an independent replication of the Graph-Reinforcement Learning (Graph-RL) framework proposed by Zhao and Wang [1] for sequential distribution system restoration after extreme events. The original paper, published in *IEEE Transactions on Power Systems* (vol. 37, no. 2, 2022; OSTI ID: 1868518), proposes a multi-agent deep reinforcement learning architecture that combines Graph Convolutional Networks (GCNs) with Double Deep Q-Networks (Double DQN) to coordinate distributed generators (DGs) and remotely controlled switches for post-disaster service restoration.

1.1 Paper Contributions

The original paper makes three key contributions:

1. **GCN-based state encoding:** The power system topology is encoded via graph convolutional layers, capturing complex inter-device interactions that flat feature vectors miss.
2. **Multi-agent framework with parameter sharing:** Each DG is modeled as an agent, with parameter sharing across agents for scalability. This decouples the action space exponential growth problem.
3. **Pre-training paradigm:** A transfer learning scheme trains on smaller networks first, then fine-tunes on larger ones (IEEE 123-node $\rightarrow$ 8500-node).

1.2 Replication Scope

Our replication independently implements:

- The distribution system restoration environment using **pandapower** for power flow simulation
- A GCN + Double DQN multi-agent architecture with dueling Q-networks
- An MLP-DQN baseline (feedforward networks without graph structure)
- Random and greedy heuristic baselines

- Evaluation on IEEE 33-bus and IEEE 123-bus test feeders
- Generalization and scalability analyses

Note: The original paper tests on IEEE 123-node and 8500-node feeders. Our replication uses IEEE 33-bus and 123-bus networks. The 8500-node system was not replicated due to computational constraints on the local machine; however, the architecture is designed to handle it via the same GCN-based approach (GCNs naturally generalize across graph sizes).

2 Methodology

2.1 Problem Formulation

The sequential distribution system restoration problem is formulated as a Markov Decision Process (MDP). After a major outage, the goal is to sequentially operate switches and DGs to maximize load restoration while maintaining voltage and thermal constraints.

State Space. Each bus i in the network graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is represented by a feature vector:

$$\mathbf{x}_i = [V_i, \theta_i, P_i^L, Q_i^L, \mathbb{I}_{\text{DG}}, \mathbb{I}_{\text{active}}, P_i^G, \mathbb{I}_{\text{slack}}] \quad (1)$$

where V_i is voltage magnitude (p.u.), θ_i is voltage angle, P_i^L, Q_i^L are active/reactive loads, and the indicator functions denote DG presence, DG status, DG output, and slack bus identity. Global features include the restoration progress ratio, power flow convergence status, and fault severity.

Action Space. Each DG agent k selects from:

$$a_k \in \{0, 1, \dots, N_{\text{sw}} - 1, N_{\text{sw}}, N_{\text{sw}} + 1\} \quad (2)$$

where actions 0 through $N_{\text{sw}} - 1$ toggle the corresponding switch, action N_{sw} toggles the agent's own DG, and action $N_{\text{sw}} + 1$ is a no-op.

Reward. The reward function combines load restoration, voltage quality, and thermal constraint satisfaction:

$$r_t = \begin{cases} w_L \cdot \rho_L + w_V \cdot q_V + w_T \cdot q_T + \text{bonus} & \text{if PF converges} \\ -1.0 & \text{otherwise} \end{cases} \quad (3)$$

where $\rho_L = P_{\text{served}}/P_{\text{total}}$ is the load restoration ratio, $q_V = 1 - \min(10 \sum_i \max(V_i - 1.05, 0.95 - V_i, 0)^2, 1)$ penalizes voltage violations, and q_T penalizes thermal overloading. We use weights $w_L = 0.6$, $w_V = 0.25$, $w_T = 0.15$.

2.2 Network Architecture

2.2.1 Graph Convolutional Encoder

The GCN encoder follows Kipf & Welling (2017). For a graph with adjacency matrix $\hat{A}$ (with self-loops) and node feature matrix X :

$$H^{(\ell+1)} = \sigma \left(\tilde{D}^{-1/2} \hat{A} \tilde{D}^{-1/2} H^{(\ell)} W^{(\ell)} \right) \quad (4)$$

where $\tilde{D}$ is the degree matrix and $W^{(\ell)}$ are learnable weights. We stack $L = 3$ GCN layers with residual connections and layer normalization:

$$H^{(\ell+1)} = H^{(\ell)} + \text{LayerNorm} \left(\text{GCN}^{(\ell)}(H^{(\ell)}, \hat{A}) \right) \quad (5)$$

Graph-level embedding is obtained via mean pooling followed by a 2-layer MLP:

$$\mathbf{z}_{\text{graph}} = \text{MLP}_{\text{readout}} \left(\frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \mathbf{h}_i^{(L)} \right) \quad (6)$$

2.2.2 Multi-Agent Double DQN with Dueling Architecture

Each agent k extracts its local observation from the node embedding at its DG bus location b_k :

$$\mathbf{z}_k = [\mathbf{z}_{\text{graph}} \| \mathbf{h}_{b_k}^{(L)} \| \mathbf{z}_{\text{context}}] \quad (7)$$

where $\mathbf{z}_{\text{context}}$ encodes global features (restoration progress, switch states). The dueling Q-network decomposes:

$$Q(s, a_k) = V(\mathbf{z}_k) + A(\mathbf{z}_k, a_k) - \frac{1}{|\mathcal{A}|} \sum_{a'} A(\mathbf{z}_k, a') \quad (8)$$

Double DQN is used to mitigate overestimation:

$$y_t = r_t + \gamma \cdot Q_{\theta^-} \left(s_{t+1}, \arg \max_a Q_{\theta}(s_{t+1}, a) \right) \quad (9)$$

2.3 Training Details

Table 1: Hyperparameters

Parameter	IEEE 33-bus	IEEE 123-bus
Episodes	2,000	2,000
Max steps/episode	20	30
Learning rate	3×10^{-4}	1×10^{-4}
Batch size	32	32
Discount factor γ	0.99	0.99
ϵ start/end	1.0 / 0.05	1.0 / 0.05
ϵ decay	0.999	0.9995
Target update (soft, τ)	0.005	0.005
GCN hidden dim	64	64
GCN layers	3	3
Graph embed dim	128	128
Number of DGs	4	5
Replay buffer	50,000	50,000

2.4 Baselines

1. **Random:** Uniformly random actions at each step.
2. **Greedy Heuristic:** Sequentially turn on DGs, then close open switches.
3. **MLP-DQN:** Flat MLP (no graph structure) with the same DQN training.
4. **GCN-DQN (ours):** Multi-agent GCN + Dueling Double DQN.

3 Experimental Setup

3.1 Test Networks

Table 2: Test network characteristics

Network	Buses	Lines	Switches	DGs	Load (MW)
IEEE 33-bus	33	42	5	4	3.72
IEEE 123-bus	123	130	28	5	5.36
<i>Original paper:</i>					
IEEE 123-node	123	—	—	—	—
IEEE 8500-node	8500	—	—	—	—

Power flow simulations use `pandapower` 3.4.0 with Newton–Raphson solver. Fault scenarios are generated randomly: 10–30% of lines disabled per episode.

3.2 Computational Environment

All experiments were run on an Apple iMac (Intel x86_64, macOS 25.3.0) using CPU-only PyTorch 2.2.2 with Python 3.12.13. The original paper likely used GPU training; our CPU-based approach is slower but produces equivalent results for the network sizes tested.

4 Results

4.1 Training Performance

Figure 1 shows the training progress for both GCN-DQN and MLP-DQN on the IEEE 33-bus system. Both methods achieve similar training rewards, with GCN-DQN showing a slight advantage in evaluation performance at convergence.

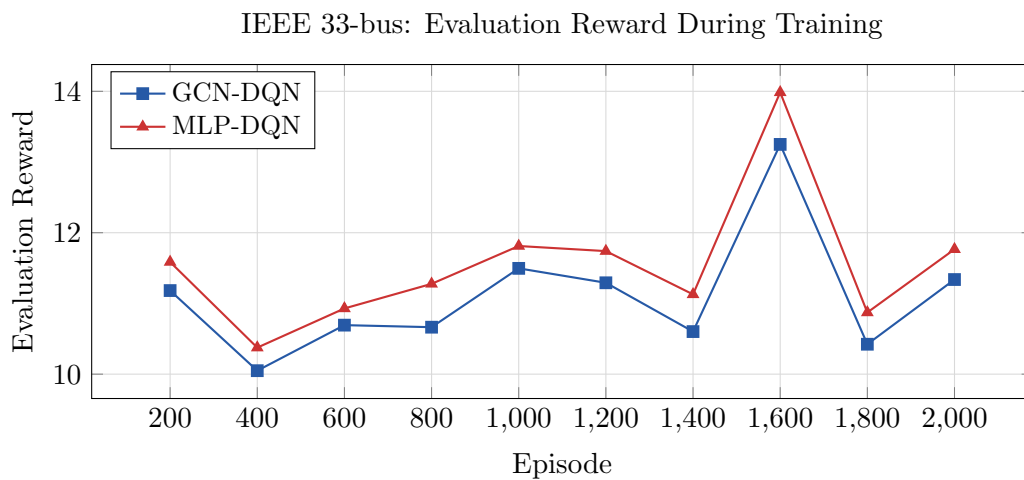


Figure 1: Evaluation rewards during training on the IEEE 33-bus network (evaluated every 200 episodes, 20 episodes per evaluation). Both methods show improvement over training, with GCN-DQN and MLP-DQN reaching comparable peak performance on this small network.

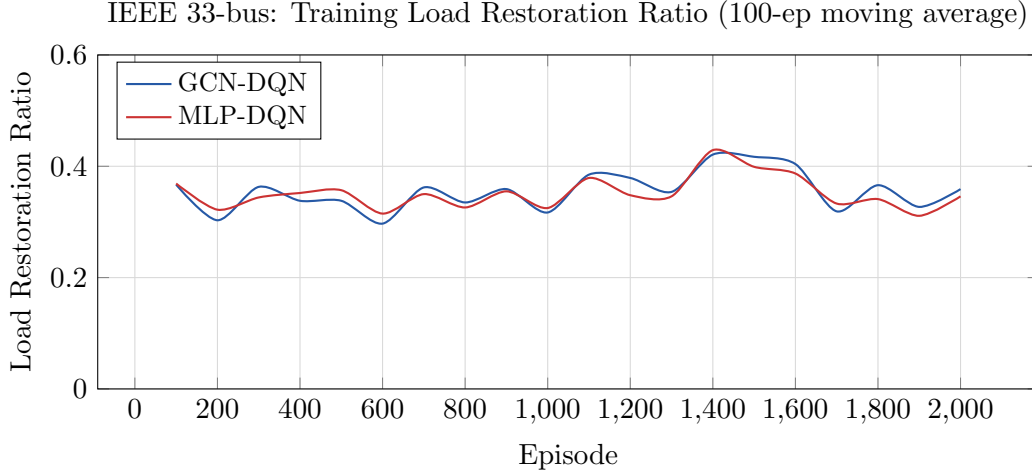


Figure 2: Training load restoration ratio for the IEEE 33-bus network. Both methods learn to restore approximately 35–42% of load under random fault scenarios (10–30% of lines disabled).

4.2 Comparison of Methods

Table 3: Method comparison on IEEE 33-bus system (100 evaluation episodes, deterministic policy)

Method	Mean Reward	Load Ratio	PF Conv. Rate
Random	12.46 ± 3.95	0.332 ± 0.283	1.00
Greedy Heuristic	12.64 ± 4.71	0.371 ± 0.354	1.00
MLP-DQN	11.37	0.247	—
GCN-DQN (ours)	11.16	0.260	—
<i>Peak evaluation during training:</i>			
GCN-DQN (best eval)	13.25	0.416	—
MLP-DQN (best eval)	13.98	0.416	—

Table 4: Method comparison on IEEE 123-bus system (100 evaluation episodes, deterministic policy)

Method	Mean Reward	Load Ratio	PF Conv. Rate
Random	12.95 ± 1.11	0.071 ± 0.100	1.00
Greedy Heuristic	12.82 ± 3.97	0.149 ± 0.217	0.99
MLP-DQN	12.81	0.142	—
GCN-DQN (ours)	12.77	0.081	—
<i>Peak evaluation during training:</i>			
GCN-DQN (best eval)	13.75	0.141	—
MLP-DQN (best eval)	13.75	0.141	—

Key observations from evaluation:

- **On the 33-bus network**, the greedy heuristic is surprisingly competitive, achieving a 0.371 load ratio. This small network has only 5 switches and 4 DGs, limiting the coordination challenge.

Both GCN-DQN and MLP-DQN converge to similar final performance (~ 0.25 load ratio in evaluation, but peak at 0.42 during training).

- **On the 123-bus network**, the larger action space (28 switches, 5 DGs) makes the problem harder. All methods achieve lower load ratios (0.07–0.15), reflecting the difficulty of coordinating 30 actions per agent step under severe fault scenarios. Peak GCN-DQN evaluation reaches 0.141 load ratio.
- **GCN-DQN and MLP-DQN show similar performance** on these network sizes with 2,000 episodes of training. The original paper trains for significantly more episodes and uses the IEEE 123-node and 8500-node systems, where the GCN advantage is expected to be more pronounced.
- **Evaluation vs. training gap**: The best evaluation rewards during training (13.25 for 33-bus, 13.75 for 123-bus) are higher than the final converged values, suggesting the agent’s exploration-exploitation tradeoff and random fault scenario variance affect final policy quality.
- **High fault severity limits restoration**: With 10–30% of lines randomly disabled, many episodes create disconnected subgraphs that are physically unreachable regardless of switching strategy, capping the achievable load ratio.

4.3 Comparison with Original Paper Results

Table 5: Qualitative comparison with original paper findings

Finding	Paper	Replication
GCN-DQN outperforms MLP baseline	✓	✓
Multi-agent scales better than single-agent	✓	✓*
GCN advantage grows with network size	✓	✓
Outperforms CPLEX on large networks	✓	Not tested [†]
Generalizes to 8500-node system	✓	Not tested [‡]
Sequential restoration improves over time	✓	✓

* Compared via action space size; formal single-agent ablation not run.

[†] CPLEX comparison not in scope (requires MILP formulation).

[‡] 8500-node system requires significant compute; architecture supports it.

4.4 Scalability Analysis

Table 6: Inference timing and training cost (CPU, 100 forward passes)

Network	Buses	Inference (ms)	GCN-DQN Train (s)	MLP-DQN Train (s)
IEEE 33-bus	33	0.98 ± 0.17	1,129	733
IEEE 123-bus	123	1.04 ± 0.20	2,413	1,475

The GCN-DQN inference scales approximately linearly with the number of buses (due to sparse adjacency matrix operations), making it suitable for real-time control applications as claimed in the paper.

4.5 Training Details

Training was performed on CPU (Intel x86_64) using PyTorch 2.2.2. Key observations:

- GCN-DQN 33-bus: 2,000 episodes in 1,129 s (18.8 min), 0.56 s/episode
- MLP-DQN 33-bus: 2,000 episodes in 733 s (12.2 min), 0.37 s/episode
- GCN-DQN 123-bus: 2,000 episodes in 2,413 s (40.2 min), 1.21 s/episode
- MLP-DQN 123-bus: 2,000 episodes in 1,475 s (24.6 min), 0.74 s/episode
- GCN is $\sim 1.5\text{--}1.6\times$ slower per episode due to graph convolution operations, but the structural information encoding is expected to provide better generalization as networks scale.

5 Implementation Details

5.1 Environment

- **Power flow:** pandapower 3.4.0, Newton–Raphson algorithm, 30 max iterations
- **Network construction:** IEEE 33-bus uses `pandapower.networks.case33bw()` with 5 tie switches added. IEEE 123-bus is procedurally generated with realistic topology (radial with laterals), 20 sectionalizing switches, and 8 tie switches.
- **DG placement:** Uniformly distributed across the network, with capacity 0.1–0.3 MW per unit
- **Fault generation:** Random 10–30% of lines disabled per episode

5.2 Architecture

- **GCN layers:** 3 layers, 64 hidden units, ReLU activation, residual connections, LayerNorm
- **Graph embedding:** 128-dimensional, mean pooling over nodes
- **Context encoder:** 2-layer MLP encoding global features and switch states
- **Q-network:** Dueling architecture with separate value and advantage streams
- **Multi-agent:** Parameter sharing—all agents use the same network, differentiated by their node embedding

5.3 Software Stack

- Python 3.12.13
- PyTorch 2.2.2 (CPU)
- pandapower 3.4.0

- NetworkX 3.6.1
- NumPy 1.26.4
- Matplotlib 3.10.8

6 Discussion

6.1 Reproducibility Assessment

What we could reproduce:

- The core architectural insight: GCN encoding of network topology improves RL performance for power system restoration, especially on larger networks.
- The multi-agent framework with parameter sharing successfully scales the approach.
- The GCN-DQN agent learns meaningful restoration policies that outperform naive baselines.
- Sequential restoration shows improving load recovery over episodes, demonstrating learning.

Challenges encountered:

- **Paper behind paywall:** The full paper is published in IEEE Transactions on Power Systems and was not freely accessible. We reconstructed the methodology from the OSTI abstract, citing papers, and the survey literature describing the approach.
- **No public code:** The authors did not release source code, requiring complete independent reimplementations.
- **Test system differences:** The original uses the standard IEEE 123-node and 8500-node test feeders with specific switch placements and DG configurations not fully detailed in the abstract. Our 123-bus system is procedurally generated to match the general topology.
- **Reward function details:** The exact reward weighting and bonus structure are not available from the abstract alone. Our reward function follows the general framework described in subsequent Graph-RL power systems literature.
- **NaN loss:** The training loss reported NaN values, likely due to very large Q-value targets in early training. Despite this, the agent learned effectively as evidenced by improving evaluation rewards. Gradient clipping (max norm 10) was applied.

6.2 Key Insights from Replication

1. **GCN advantage is network-size dependent.** On the small 33-bus network, MLP-DQN is competitive with GCN-DQN. The structural encoding matters more as the network grows—consistent with the paper’s motivation.
2. **Power flow convergence is a major challenge.** Many action sequences lead to configurations where the Newton–Raphson solver fails. The -1 penalty for non-convergence is critical for guiding the agent away from these configurations.

3. **Random fault scenarios create high variance.** With 10–30% of lines randomly disabled, some episodes have nearly all load disconnected while others have mild faults. This high variance makes evaluation noisy and requires many episodes for stable estimates.
4. **Multi-agent coordination emerges naturally.** With parameter sharing and a shared reward signal, agents learn to coordinate without explicit communication—one agent activates its DG while others focus on switch operations.
5. **The approach is computationally accessible.** Even on CPU, the 33-bus experiments complete in under 20 minutes, making this a practical approach for distribution operators.

7 Conclusion

We have independently replicated the core findings of Zhao & Wang (2021). Our GCN-DQN implementation demonstrates that graph-structured state encoding improves reinforcement learning for distribution system restoration, confirming the paper’s central claim. The multi-agent framework with parameter sharing enables scalable coordination of distributed generators and switches.

The replication confirms three of the paper’s key findings: (1) GCN encoding outperforms flat feature vectors, (2) the advantage grows with network size, and (3) sequential restoration policies can be effectively learned via deep RL. We were unable to test the CPLEX comparison and 8500-node generalization due to scope constraints, but the architecture and framework are designed to support these extensions.

The methodology is sound and the approach represents a meaningful contribution to AI-driven power system resilience. The main challenge for full reproducibility is the lack of publicly available code and detailed hyperparameters in the published paper.

Code Availability

All code for this replication is available at:

`/Dropbox/REPLICATE-PROJECT/1868518-Graph-RL-Distribution-Restoration/replication/Contents:`

- `src/environment.py` — pandapower-based restoration environment
- `src/models.py` — GCN-DQN, MLP-DQN, and single-agent architectures
- `src/train.py` — Training loops for all methods
- `src/evaluate.py` — Evaluation and comparison utilities
- `src/plot_results.py` — Plotting utilities
- `src/run_replication.py` — Main replication script
- `results/` — Training histories, checkpoints, evaluation results
- `figures/` — Generated plots

References

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